

Birehan Zewdie

✉ babianteneh4@gmail.com  birehan  birehan  birehananteneh4

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, *The Cool Company*

May 2025 – Jan 2026 | New York, USA

- Built a data-driven AI floor-price prediction system, leveraging machine learning with LightGBM to dynamically optimize pricing per ad request for DSPs, resulting in a 7% increase in revenue.
- Built a client revenue-forecasting system that predicts daily revenue using scraped domain data, Google PageSpeed metrics, WHOIS information, historical performance data, and LLM-generated features, achieving 86%+ prediction accuracy.

Machine Learning Engineer, *Adludio*

May 2024 – May 2025 | London, UK

- Optimized campaign ROI by implementing an automated pacing algorithm leveraging predictive modeling and real-time bid adjustments, resulting in a 16% improvement in KPIs.
- Enhanced ad performance by developing a creative scoring system using historical data and LLMs to evaluate creative variations and identify high-engagement assets, leading to a 10% lift in engagement rate and a 5% improvement in CTR and CPA.
- Accelerated creative production by creating an LLM-driven pipeline that generates on-brand ad concepts aligned with campaign objectives, reducing ideation time by 50%.
- Increased conversion rates and profit margins by developing an inventory scoring system using DCN v2 and LTR algorithms to optimize bid adjustments across publishers, formats, and devices, resulting in a 15% uplift in conversions and a 10% increase in profit margins.

AI Software Developer | Team Lead, *Eskalate S.C.*

Mar 2023 – Apr 2024
Addis Ababa, Ethiopia

- Led a cross-functional team to successfully launch HakimHub, Ethiopia's first AI-powered medical recommendation platform, leveraging Jira for structured project planning, task tracking, and seamless team coordination throughout the development lifecycle.
- Fine-tuned a Llama 3-based chatbot that employs AI agents for personalized symptom assessments and follow-ups, integrating a database tool to query doctors and hospitals.

Software Engineer, *AiQEM Tech*

Jul 2023 – Sep 2023
Addis Ababa, Ethiopia

- Developed a React-based dashboard for campaign statistics visualization by designing and implementing a dashboard using React to display key campaign metrics, such as engagement rates and trends over time.
- Collaborated with teams to enhance dashboard functionality by working with cross-functional teams to gather requirements and improve the dashboard's usability for advertisers.

SKILLS

- **Languages:** Python, Java, C++, JavaScript, C#, SQL, Dart, HTML/CSS, JavaScript, TypeScript
- **Technologies:** TensorFlow, PyTorch, scikit-learn, Airflow, Spark, LangChain, Huggingface, Git, Docker, AWS
- **Concepts:** ML Algorithms, Model Evaluation, Feature Engineering, Data Preprocessing, MLOps

PROJECTS

Storyboard Synthesis for Digital Advertising

Feb 2024 – Feb 2024

- Created a machine learning solution that automated storyboard making, cutting production time from a week to just a few hours for digital ads, allowing quicker delivery to clients in a competitive market.
- Conducted exploratory data analysis and optimized YOLO object detection, while exploring image generation techniques to enhance storyboard quality and alignment.

Automated Prompt Engineering for RAG Systems

Jan 2024 – Jan 2024

- Implemented automated prompt generation, evaluation data creation, and testing tools to optimize Language Models (LLMs) in RAG systems.
- Integrated vector database to establish contextual relevance, enabling efficient prompt engineering within the LangChain framework.

Email Classification with Fine-Tuned Transformers	Jan 2025 – Feb 2025
- Fine-tuned BERT, RoBERTa, and DistilledBERT on 100K+ emails to achieve 93% accuracy. - Employed quantization (FP16, INT8) and ONNX conversion for seamless mobile deployment. Tech stack: Hugging Face Transformers, PyTorch, ONNX, and quantization libraries.	
Loan Risk Prediction Machine Learning Pipeline	Sep 2024 – Sep 2024
- Built a loan risk pipeline with data ingestion, LightGBM, and FLAML (83.78% accuracy). - Deployed with MLflow & Docker for REST predictions and CI/CD retraining. Tech stack: Python, LightGBM, FLAML, MLflow, Docker, Github Actions.	
Legal Expert Retrieval-Augmented Generation (RAG)	Feb 2024 – Feb 2024
- Created a Contract Q&A system for Lizzy AI using RAG for legal analysis. - Achieved 87% relevance in contract analysis through semantic chunking and hybrid retrieval. Tech stack: Python, Hugging Face, Weaviate, and LangChain, Fast API, React	

EDUCATION	
BSc. in Software Engineering (AI Stream) , <i>Addis Ababa University</i>	Addis Ababa, Ethiopia
OOP, Database, Software Development, Machine Learning, and NLP, Reinforcement Learning.	
AI and Machine Learning Intensive Training , <i>10 Academy</i>	Santa Clara, USA
software engineering program , <i>A2SV backed by Google</i>	Palo Alto, USA

CERTIFICATES AND AWARDS	
AWS Certified Machine Learning Engineer Associate Machine Learning A-Z: AI, Python Winner, DevFest 2022 Hackathon (Google Developers) Machine Learning with Python The Complete Web Development EDA for Machine Learning	